

# CELIA RUBIO-MADRIGAL

 celrm.github.io

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I am a PhD candidate at CISPA, where I study challenges in graph learning regarding the dual role of graphs as data and computation structures. I hold degrees in Mathematics and Computer Science from Universidad Complutense de Madrid, and a "la Caixa" Foundation Fellowship that funded my master's at Strathclyde, where I earned the Best Overall Prize. I am also part of IT for the European Girls' Mathematical Olympiad every year.

## EXPERIENCE



- **Doctoral researcher** – CISPA Helmholtz Center for Information Security 2023-now  
– Relational Machine Learning Lab  – Supervisor: Dr. Rebekka Burkholz 
- **RA: Departmental collaboration** – Ministerio de Educación & Universidad Complutense de Madrid 2021-22  
– Identification of Boolean functions that need big circuits  – Supervisor: Dr. Ismael Rodríguez Laguna 
- **REU: Introduction to research ICMAT-Severo Ochoa** – Instituto de Ciencias Matemáticas (ICMAT) 2021  
– Differential geometry methods in ML applications  – Supervisor: Dr. David Martín de Diego 
- **Consultant Developer intern** – Guidewire Software – Node.js search engine for internal raised issues 2021

## EDUCATION



- **PhD in Computer Science** – Universität des Saarlandes (Saarbrücken, Germany) 2023-now
- **MSc in Advanced CS with AI** – University of Strathclyde (Glasgow, UK) – 8.57/10 (Distinction) 2022-23  
– Fully funded by "la Caixa" Foundation Fellowship  – Earned the Department's Best Overall Prize.
- **BSc in Mathematics** – Universidad Complutense de Madrid (Spain) – 9.08/10 2017-22
- **BSc in Computer Science** – Universidad Complutense de Madrid (Spain) – 9.16/10 2017-22
- **Professional Music Degree in Baroque Recorder** – Conservatorio Profesional de Música Arturo Soria 2006-16

## PUBLICATIONS



- **Peer-reviewed publications** \* Equal contribution.  
– Fixed Aggregation Features can rival GNNs. C.R.M., R. Burkholz.  
20th Women in Machine Learning Workshop (WiML) at 39th NeurIPS, 2025.
- GNNs getting ComFy: Community and feature similarity guided rewiring. C.R.M.\*, A. Jamadandi\*, R. Burkholz.  
13th International Conference on Learning Representations (ICLR), 2025. 
- Spectral graph pruning against over-squashing and over-smoothing. A. Jamadandi\*, C.R.M.\*, R. Burkholz.  
38th Conference on Neural Information Processing Systems (NeurIPS), 2024. 

- **Under review**

- *Boosting for predictive sufficiency*. A.G. Reddy, R. Verma, [C.R.M.](#), R. Burkholz, K. Muandet. 2025.
- *When shift happens-Confounder is to blame*. A.G. Reddy, [C.R.M.](#), R. Burkholz, K. Muandet. arXiv, 2025. 
- *HAM: A hyperbolic step to regulate implicit bias*. T. Jacobs, A. Gadhikar, [C.R.M.](#), R. Burkholz. arXiv, 2025. 

- **Preprints, theses, and columns**

- *Rendering string diagrams recursively*. [C.R.M.](#), J. Hedges. arXiv, 2024.   
Based on Master's thesis at University of Strathclyde, 2023.
- *Analysis of a neural network for the identification of complex Boolean functions*. [C.R.M.](#)  
Bachelor's thesis at Universidad Complutense de Madrid, 2022. 
- *akesi li kama*. [C.R.M.](#), Sci-Fdl: Revista de Ciencia Ficción, Vol. 27, ISSN 1989-8363, 2022. 
- *Serialismo y matemáticas (I)* , *(II)* , *(III)* , *(IV)* . [C.R.M.](#), DivulgaMAT, Música y Matemáticas, 2019.

## TALKS



- **Invited talks**

-  Invited talk: Graph Learning on Wednesdays (GLOW) community: *online* Oct 2025
-  Visit & invited talk: Seminar of the TUWien ML Research Unit: *Vienna, Austria* May 2025
-  Invited talk: Cohere Labs / Cohere For AI - Open Science Community: *online*  Feb 2025

- **Contributed talks**

-  Lightning talk: Workshop on Geometry, Topology, and ML (GTML): *Leipzig, Germany* Nov 2025
-  Two lightning talks: Workshop on Mining and Learning with Graphs (MLG): *Porto, Portugal* Sep 2025
-  Lightning talk: International Conference on Network Science (NetSci): *Maastricht, Netherlands* Jun 2025
-  Contributed talk: Helmholtz AI Conference: AI for Science (HAICON): *Düsseldorf, Germany* Jun 2024

## AWARDS



- **Postgraduate**

- Best Overall Performance Prize: *University of Strathclyde's top student in MSc Advanced CS* 2023
-  Nova III Student List: *Spain's top 10 talents under 25 in Computer Science* 2023
- 9th Heidelberg Laureate Forum, with interviews:  HLFF Spotlight,  Tribuna Complutense 2022
-  "la Caixa" Foundation Fellowship, with interviews:  El País,  El Plural,  TeleMadrid 2022

- **Undergraduate**

-  Aula de Verano "Blas Cabrera" for Spain's top undergrads: *Intl. Univ. Menéndez Pelayo (UIMP)* 2022
-  SWERC contestant (competitive programming): *ICPC Southwestern Europe Regional Contest in Italy* 2022
- Becas de Excelencia: *Excellence Grants for undergrads with >9/10 in Madrid (2100€)* 2018-22

- **Pre-undergraduate**

-  EGMO contestant representing Spain: *European Girls' Mathematical Olympiad in Switzerland* 2017
-  Premio Extraordinario de Bachillerato: *Madrid's top 25 in general-content exam (1000€)* 2017
-  Leading Program "la Caixa": *Madrid's top 20 university admission scores* 2017
- 10/10 with Honours at IES San Mateo: *Top 5 students at Spain's only public High School of Excellence* 2017
- ESTALMAT: *2-year mathematics talent program for top 25 high school students in Madrid* 2012-14

## TEACHING



- **Master's and Bachelor's thesis advising** – *CISPA & Universität des Saarlandes* **2024-now**
  - Florian Sollinger: Which atoms make a molecule toxic? A machine learning approach (BSc TBD).
  - Chinmay Joshi: The effects of data pruning on memorization (MSc TBD).
  - Adarsh Jamadandi: On the importance of graph-task alignment for graph neural networks (MSc 2024).
- **Teaching Assistance: Laboratory demonstrator (7h/week)** – *University of Strathclyde* **2023**
  - “CS310 Foundations of AI” for Year 3 BSc Computer Science – Lecturer: Dr. Andrew Abel

## SERVICE



- **Mathematical Olympiad service** – *Organizational support at international events*
  - IT at European Girls' Mathematical Olympiad (EGMO) with Joseph Samuel Myers **2023-now**
    - I process all problem translations for the Jury before the competition, manage the contest's Q&A and internal communication channels, and upload all contestants' marks during coordinations, among other tasks.
  - Deputy Leader of Spain at EGMO 2022 and PAGMO 2021.
  - Jury at the local phases of 57th to 61st Olimpiada Matemática Española (OME) Madrid.
  - Guide at the Iberoamerican Mathematical Olympiad 2018.
- **Academic service** – *Conferences and workshops* **2024-now**
  - Program Committee and reviewer for MLG 2025. – Reviewer for ICLR 2026 and CPAL 2026.
  - Volunteer at 19th and 20th Women in Machine Learning Workshop (WiML) at NeurIPS 2024 and 2025.
- **Student representation** – *Facultad de Informática, Universidad Complutense de Madrid* **2020-22**
  - Secretary of the Fdl Student Council 2021-22. – Member of the Fdl Faculty Board 2020-22.
- **L<sup>A</sup>T<sub>E</sub>X development** – *Creation of the open-source L<sup>A</sup>T<sub>E</sub>X package: ddphonism* **2019**
  - Generates 12-tone music diagrams with TikZ; included in MiKTeX and TeXLive.
- **Outreach in mathematics** – *Videos for the Spanish math education community* **2018-21**
  - Olympic role models , Science is written with an A , speech at 20th anniversary of ESTALMAT .

## LANGUAGES



Spanish – Native      English – TOEFL iBT 110/120 (2021)      German – Goethe-Zertifikat B1 (2017)